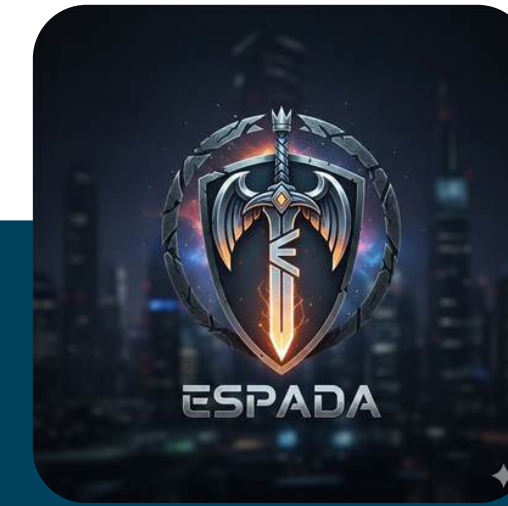




Team Name: Espada

Project Name: Smart Solar– AI-Driven Solar–Biogas System for Sustainable Communities

Track: Innovation Challenge

SDG 11 - Sustainable Cities and Communities



Team Members



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Associate Professor & Guide



Vigneshwaran J
Lead and System Architect



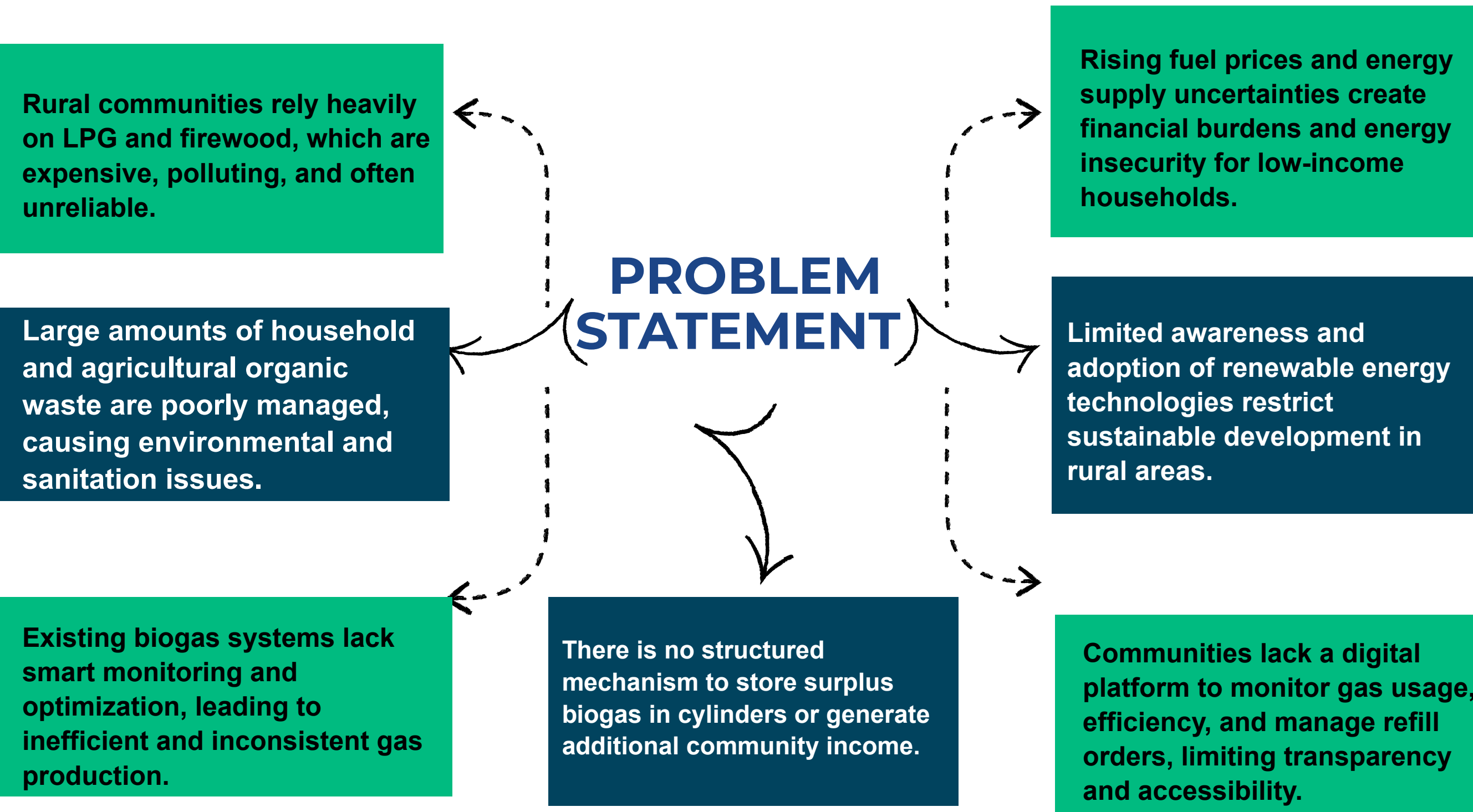
Divya Dharshini A
**Web Dashboard and
Software Lead**



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**Embedded Systems
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Hemamalar D
**Technical
Documentation &
Research Lead**



RENDERING: The Facts

Render: "To reduce, convert, or melt down (fat) by heating" from the French word "rendre", meaning "to give back"
RENDERING IS RECYCLING!

FROM SCRAPS TO VALUABLE COMMODITIES
Rendering reclaims the meat and bone scraps from animal agriculture and grocery store leftovers and transforms them into ingredients for new products. This recycles these raw materials into 29 billion pounds of fat, oil and protein.

FROM FRIES TO FUEL
Renderers recycle billions of pounds of used cooking oil each year into biodiesel and renewable diesel. Rendered protein and fats are also used to produce hundreds of industrial and personal care products.

ROUGHLY 50% OF AN ANIMAL IS CONSIDERED INEDIBLE BY AMERICANS

FEEDING THE FUTURE
Rendering helps grow the next generation of food by "recycling" unwanted meat into new and clean ingredients for animal food and fertilizer.

RENDERING RECLAIMS AND PROTECTS
3.7 billion gallons of water that would otherwise be wasted are reclaimed during rendering, cleaned, and returned to rivers and streams. Rendering also improves water quality by reducing grease and oil that clog sewer and wastewater systems.

RENDERING COMBATS CLIMATE CHANGE
Rendering protects the environment from high greenhouse gas emissions of other disposal methods. It reduces the environmental impacts of animal agriculture by sequestering 5 times more greenhouse gases than are produced.
This is equal to removing **18.5 million** cars off the road each year.
If all renderable products were sent to landfills, all available space would be gone in **4 years**.

SUSTAINING THE ECONOMY
The rendering industry is sustainable and financially stable with an annual economic contribution of **\$10 billion**

QUALITY JOBS
Rendering employs skilled workers with full-time jobs and benefits. Employee retention rates are high, contributing to more stability in local communities, many in rural America.



Introduction



1

Renewable Community Energy Solution

The system is designed to provide affordable, clean, and sustainable cooking fuel for rural and semi-urban communities.

2

Waste-to-Energy Conversion

It converts household and agricultural organic waste into methane-rich biogas through anaerobic digestion.

3

Solar-Powered Smart Monitoring

Solar energy powers sensors and AI-based control systems to optimize gas production and ensure safe operation.

4

Storage and Revenue Generation

The produced biogas is stored, distributed to households, and surplus gas can be compressed into cylinders for sale to generate income.

Research Survey



Existing Technologies

- Anaerobic digestion produces biogas
- Solar PV for powering sensors and automation
- Community biogas cuts LPG dependence
- IoT sensors improve monitoring & safety



Current Cooking Energy Methods

- LPG, firewood, & electricity dominant
- Issues: high fuel cost & air pollution
- Organic waste mostly discarded



Related Work and gaps

- Traditional biogas lacks smart control
- Solar rarely integrated with biogas
- Few combined waste-energy models
- Lack of scalable community solutions



Research Gap Identified

- No smart solar-biogas systems
- Limited decentralized gas access
- Missing real-time monitoring apps
- Need affordable, scalable solution



Interference from Survey

- Strong potential for renewable cooking
- Reduce LPG costs & dependency
- Smart monitoring boosts efficiency
- Surplus gas can generate income

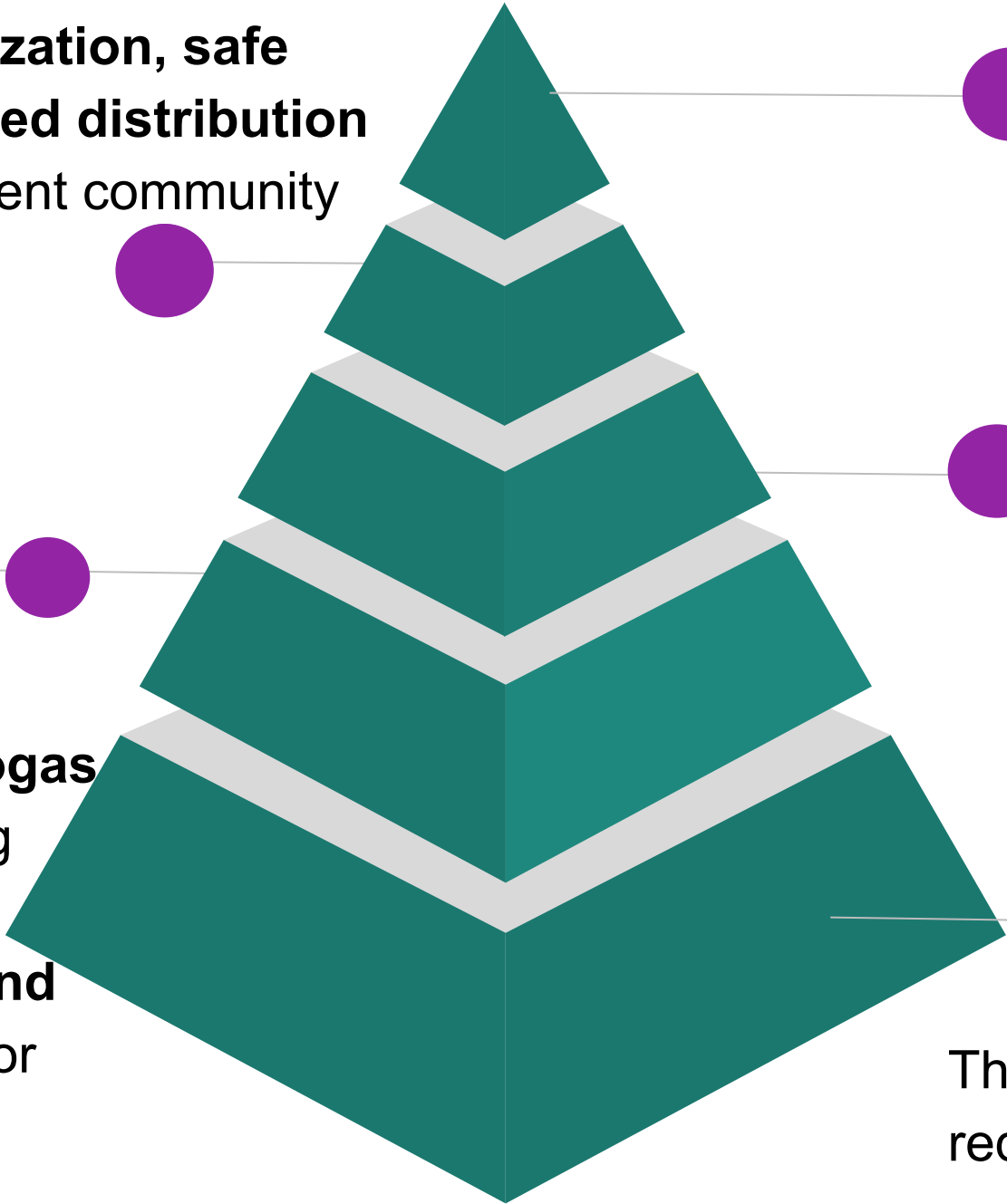


System Design Approach

The system follows a **modular and scalable design**, integrating **biogas production**, **solar-powered smart monitoring**, **AI optimization**, **safe storage**, and **app-based distribution management** for efficient community operation.

Energy Utilization Method

The system converts **organic waste into biogas for cooking** while using **solar energy to power monitoring, control, and automation systems** for efficient operation.



Proposed Solution

Proposed System Overview

The proposed system converts **organic waste into biogas**, uses **solar-powered AI monitoring for optimization**, and enables **safe storage, distribution, and app-based gas management** for communities.

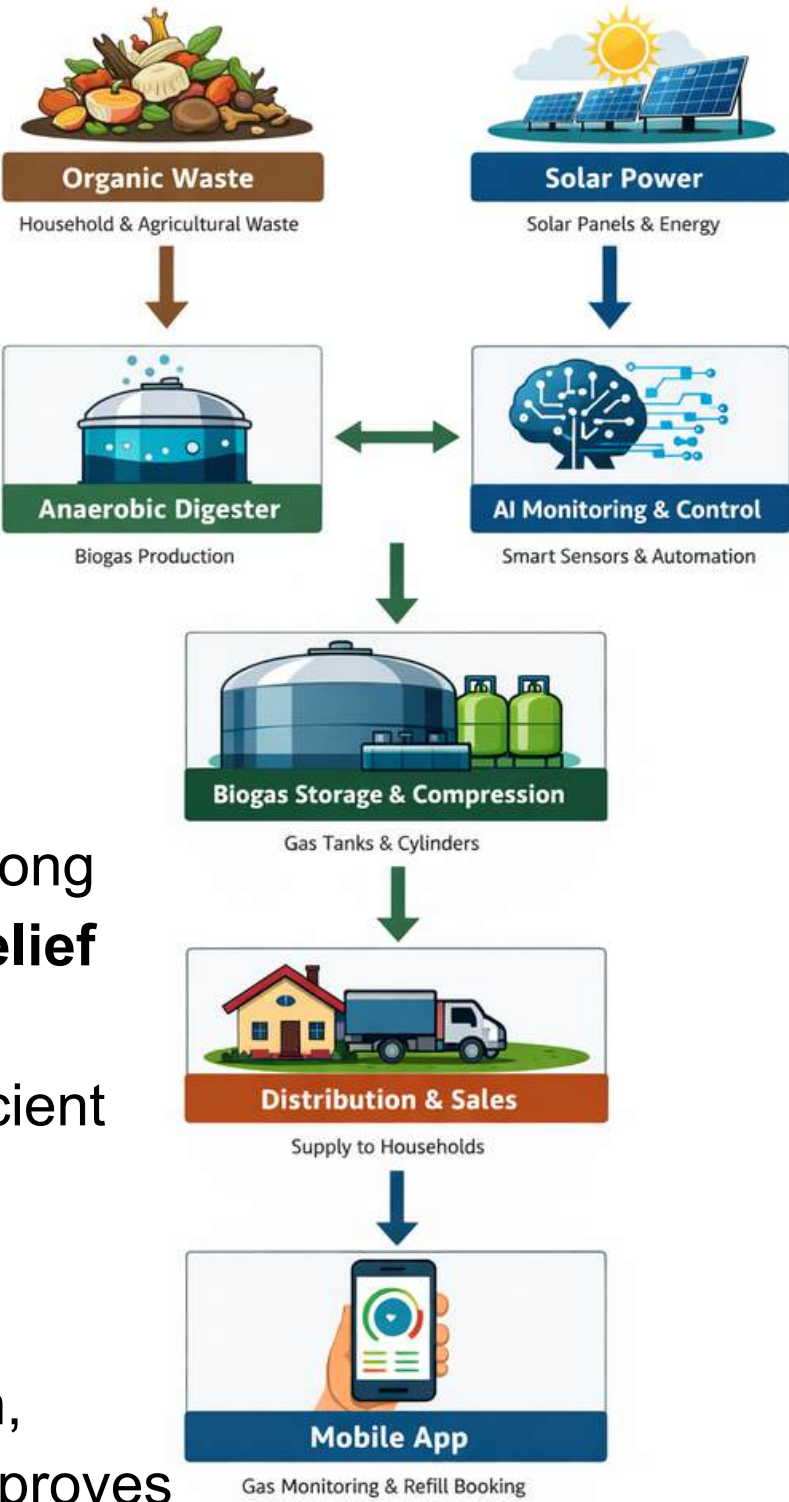
Control & Safety Mechanism

The system uses **real-time sensor monitoring and AI-based control** along with **gas leak detection, pressure relief valves, and automatic shut-off mechanisms** to ensure safe and efficient operation.

Sustainability Impact

The system promotes **clean energy adoption**, reduces **LPG use and carbon emissions**, improves **waste management**, and creates **economic benefits** for the community.

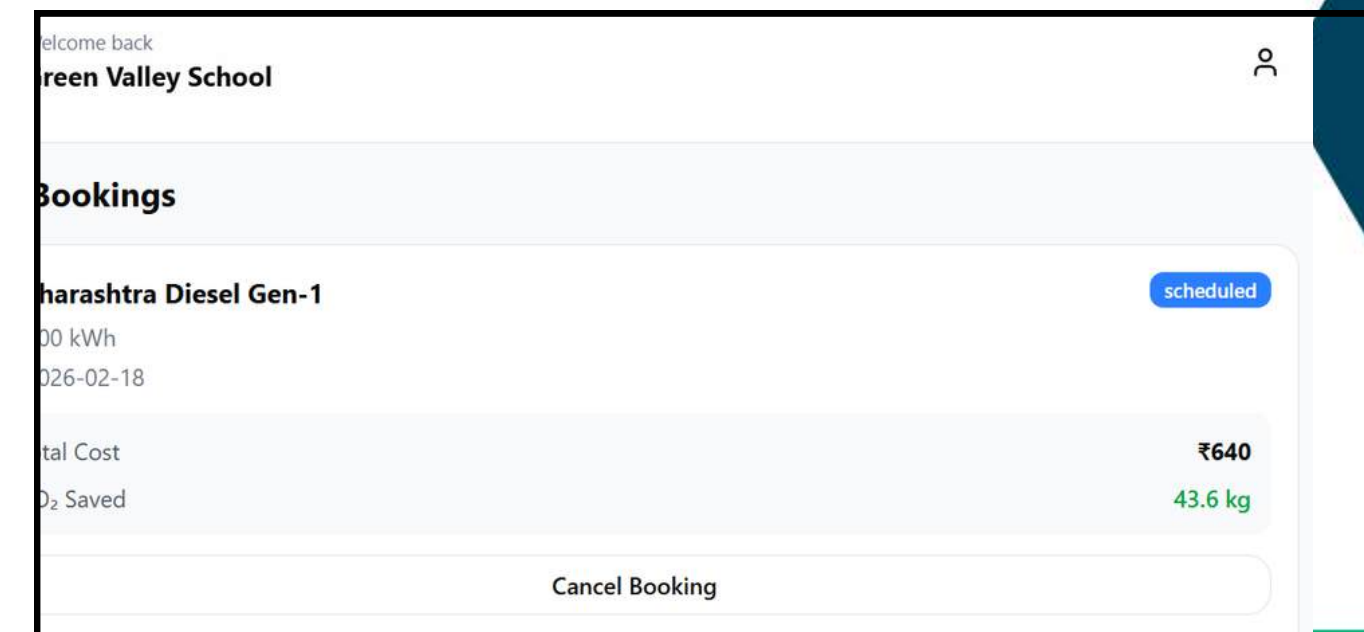
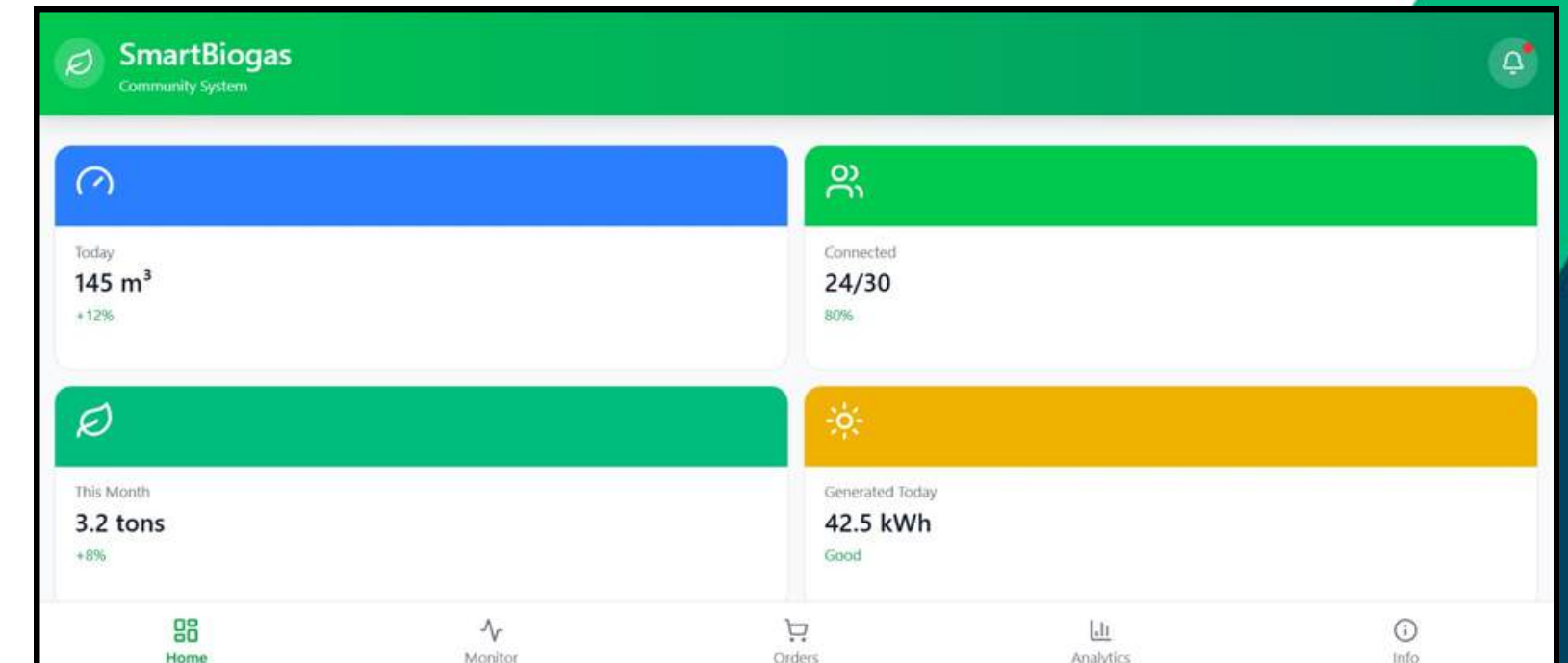
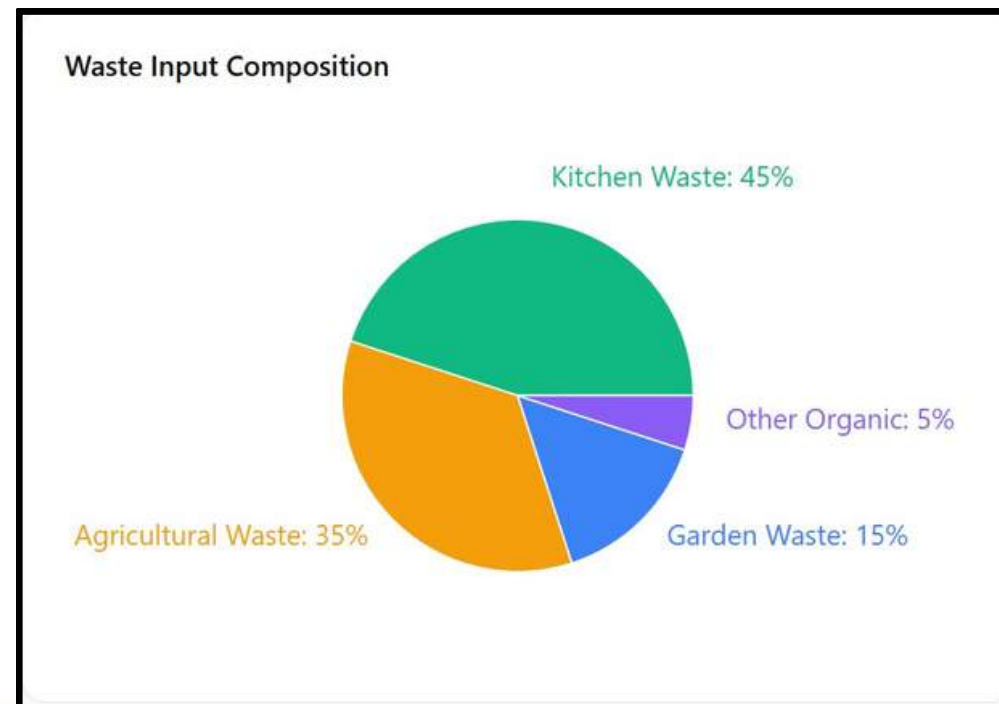
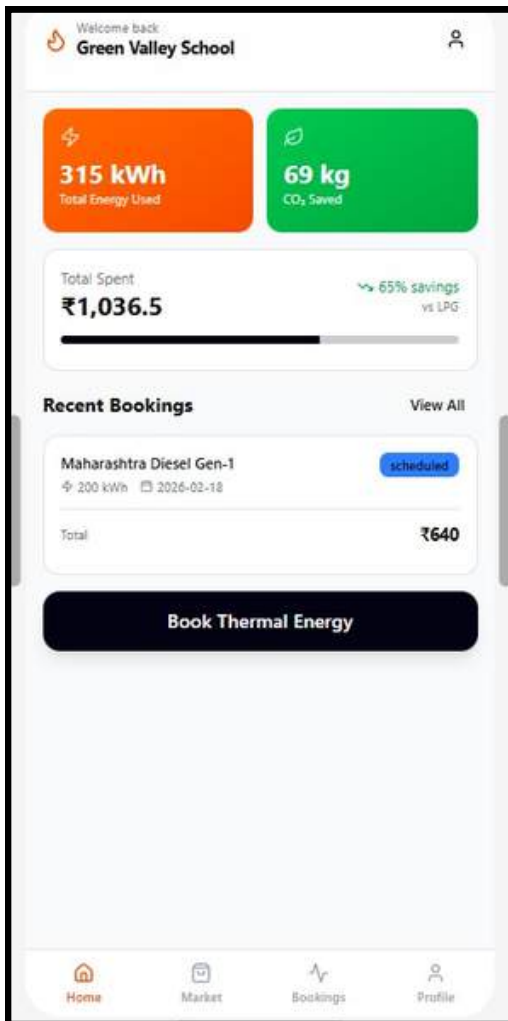
Solution Flow Chart

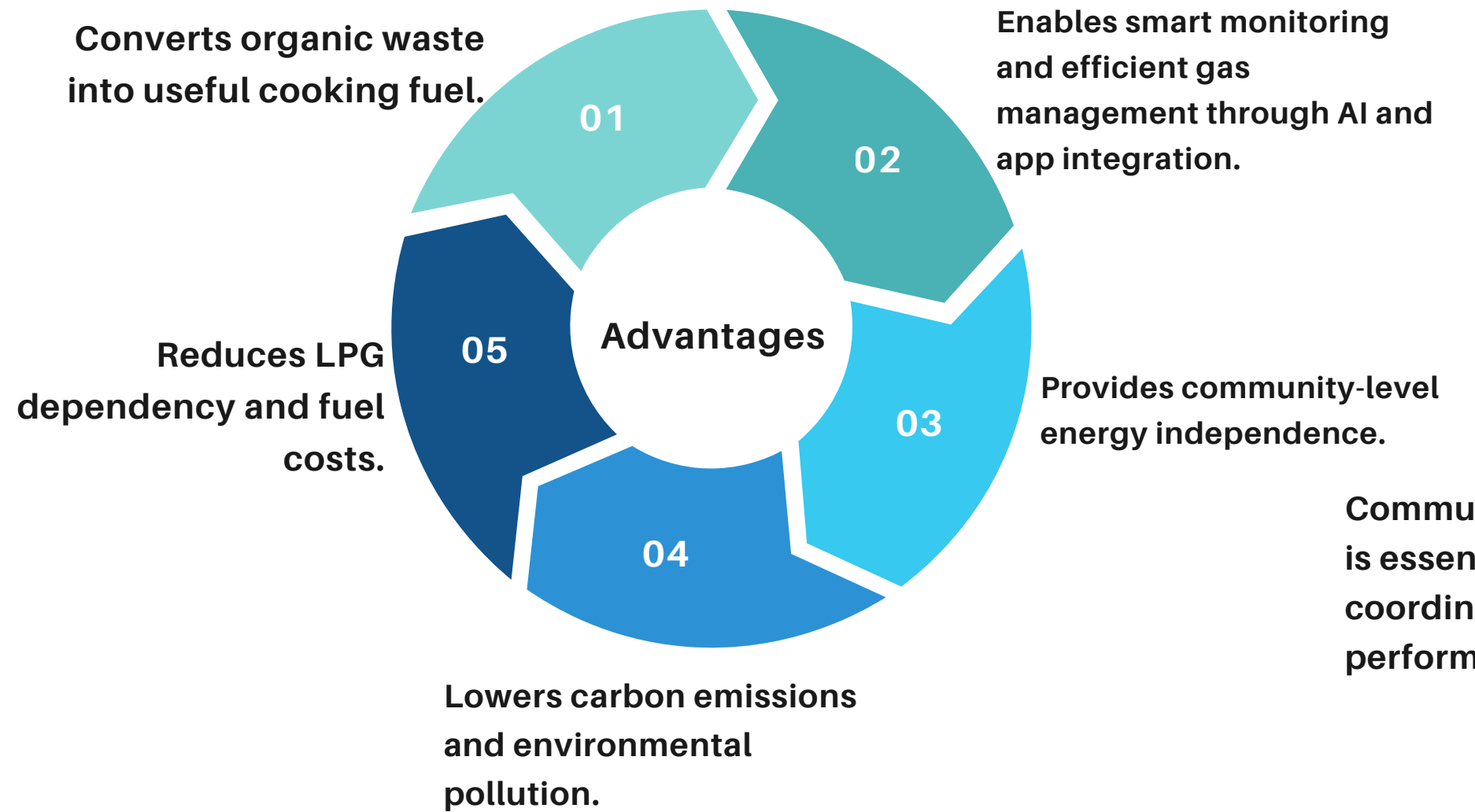


YESIST12 App Prototype



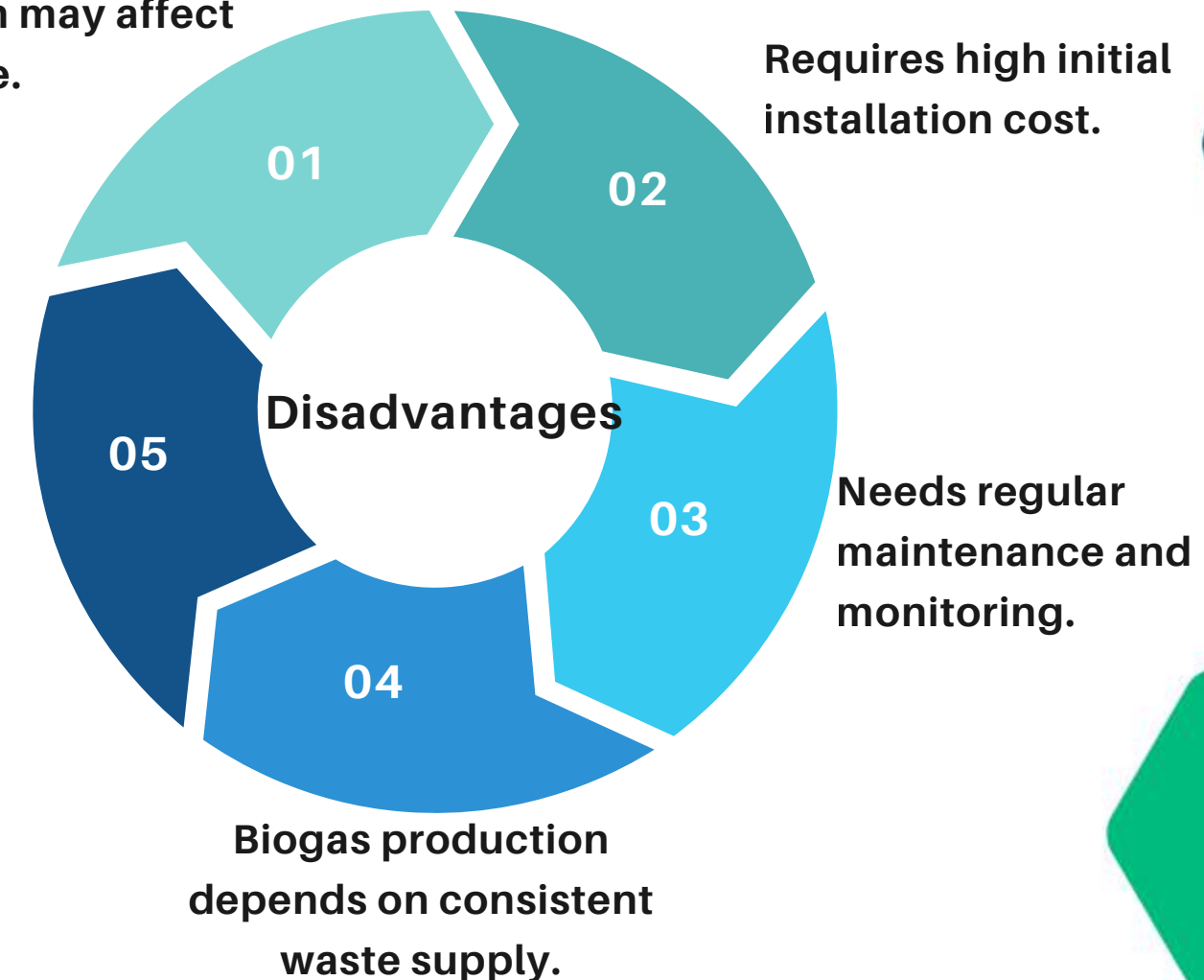
Website





Community participation is essential, and lack of coordination may affect performance.

Requires technical knowledge for AI and sensor management.



Conclusion:



Sustainable Energy Solution

The proposed system provides a renewable and eco-friendly alternative to conventional cooking fuels by converting organic waste into clean biogas.

Environmental Impact

It reduces carbon emissions, improves waste management, and promotes responsible use of renewable resources.

Economic Benefits

By lowering LPG costs and enabling surplus gas sales, the system creates financial savings and additional income opportunities for communities.

Smart and Scalable Model

With solar-powered monitoring, AI optimization, and app-based management, the system offers a modern, scalable solution for community-level clean cooking.

Community Empowerment and Energy Independence

The system empowers local communities by transforming waste into energy, strengthening self-reliance, improving energy security, and promoting long-term sustainable development.

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Thank You

